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Final Report

Date: 25.08.2017

Patient name: **Alammar, Reem Saleh Salem**

Your ref.: **067959**

DOB (dd.mm.yyyy): **14.04.1985**

Sample collection date (dd.mm.yyyy): **not available**

Order received (dd.mm.yyyy): **28.07.2017**

Sex: **female**

Patient no.: **1222757**

Sample type: **blood, EDTA**

Order no.: **62400173**

Whole Exome Sequencing (CentoXome GOLD®) was performed.

Clinical information: the proband is asymptomatic. She has 24 years old male cousin with hemophilia A which has no mutation detected. Focus on F8 gene has been requested.

Variants with possible relevance:

F9 (NM_000133.3, sequencing)	heterozygous variant c.855G>C p.(Glu285Asp)
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Please see table below for a detailed description of the detected variants

The carrier status of the hemophilia B is confirmed.

Interpretation

The F9 variant c.855G>C p.(Glu285Asp) causes an amino acid change from Glu to Asp at position 285. This variant has been confirmed by Sanger sequencing. This variant has previously been described as disease causing for haemophilia type B by Al-Allaf, 2017 (PMID: 28270892) in one affected baby boy with episodic bleeding and severe form of hemophilia B. **It is classified as variant of uncertain significance (class 3) according to the recommendations of Centogene and ACMG (please, see additional information below).**

Based on the clinical information provided, we have paid specific attention to the F8 gene. We did not detect any relevant variant in this gene. However, pathogenic variants cannot be completely excluded since not all exons were fully covered due to limitations of the method. For this gene, an overall coverage of 98.38% was achieved (>20x), with 119 missing base pairs (coding region including +/- 2bp). Please be advised that whole exome sequencing for diagnostic purposes does not provide full coverage for all genes and cannot detect large deletions/duplications.

Pathogenic variants in F9 gene are associated with hemophilia B, an X-linked disorder. Hemophilia B is a form of hemophilia characterized by spontaneous or prolonged hemorrhages due to factor IX deficiency. Hemophilia primarily affects males, but a symptomatic form of hemophilia B in female carriers has also been described with a generally milder clinical picture. People with this condition experience prolonged bleeding or oozing following an injury, surgery, or having a tooth pulled. In severe cases of hemophilia, continuous bleeding occurs after minor trauma or even in the absence of injury (spontaneous bleeding). Serious complications can result from bleeding into the joints, muscles, brain, or other internal organs. Milder forms of hemophilia do not necessarily involve spontaneous bleeding, and the condition may not become apparent until abnormal bleeding occurs following surgery or a serious injury.

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Recommendations

We recommend carrier testing for all informative family members (especially affected cousin) to establish whether the variant is associated with the disorder or not. Basic clinical information and relationship for each analyzed family member is needed for a comprehensive evaluation of the data. Genetic counselling is recommended. Genetic counselling is recommended.

Incidental findings

We did not detect any class 1 or 2 variants in the genes for which incidental findings are reported based on the ACMG guidelines.

If you have any further questions or analysis requests, please do not hesitate to contact us at support@centogene.com.

Best regards,



Prof. Peter Bauer, MD
Chief Scientific Officer Human Geneticist



Lia Abbasi Moheb, PhD
Clinical Scientist

Analysis statistics WES

Average	% target bp covered					
coverage (X)	0X	≥ 1X	≥ 5X	≥ 10X	≥ 20X	≥ 50X
127.83	0.31	99.69	99.27	98.44	95.49	79.05

CENTOGENE Mendelian variant classification (detected in the genes with established disease association)

Class 1 – Pathogenic
Class 2 – Likely pathogenic
Class 3 – Variant of uncertain significance (VUS)
Class 4 – Likely benign
Class 5 – Benign
Class 6 – Disease-associated variant

Detailed description of the detected variants

Gene	Variant Coordinates	In silico parameters*	Allele Frequencies**	Type and Classification***
F9	ChrX(GRCh37):g.138643699G>C NM_000133.3:c.855G>C p.(Glu285Asp)	PolyPhen: Benign Align-GVGD: C0 SIFT: Tolerated MutationTaster: Polymorphism Conservation: nt no/aa moderate	ExAC: - ESP: - 1000 G: - CentoMD: -	Missense Uncertain significance (class 3)

Variant description based on Alamut Batch 1.7 (latest database available). * AlignGVGD: C0: least likely to interfere with function, C65: most likely to interfere with function, splice prediction tools: SSF, MaxEnt, HSF. ** Exome Aggregation Consortium database (ExAC), Exome Sequencing Project (ESP), 1000Genome project (1000G) and CentoMD 4.0. *** based on ACMG recommendations.

Methods

RNA capture baits against approximately 60 Mb of the Human Exome (targeting >99% of regions in CCDS, RefSeq and Gencode

CLIA registration 99D2049715; CAP registration 8005167. Scientific use of these results requires permission of CENTOGENE. If you would like to download your reports from our web portal, please contact us to receive your login and password. More information is available at www.centogene.com or support@centogene.com.



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databases) is used to enrich regions of interest from fragmented genomic DNA with Agilent's SureSelect Human All Exon V6 kit. The generated library is sequenced on an Illumina platform to obtain an average coverage depth of ~100x. Typically, ~97% of the targeted bases are covered >10x. An end to end in-house bioinformatics pipeline including base calling, alignment of reads to GRCh37/hg19 genome assembly, primary filtering out of low quality reads and probable artefacts, and subsequent annotation of variants, is applied. All disease causing variants reported in HGMD®, in ClinVar or in CentoMD® as well as all variants with minor allele frequency (MAF) of less than 1% in ExAC database are considered. Evaluation is focused on coding exons along with flanking +/-20 intronic bases. All pertinent inheritance patterns are considered. In addition, provided family history and clinical information are used to evaluate eventually identified variants. All identified variants are evaluated with respect to their pathogenicity and causality, and these are categorized into classes 1 - 6 (see above). All variants related to the phenotype of the patient, except benign or likely benign variants, are reported. The F9 gene (exon 8) was analysed by PCR and sequencing of both DNA strands of the entire coding region and the highly conserved exon-intron splice junctions. The reference sequence is / sequences are: F9: NM_000133.3.

Limitations

Test results are interpreted in the context of clinical findings, family history and other laboratory data. Only variations in genes potentially related to the proband's medical condition are reported. Rare polymorphisms may lead to false negative or positive results. Misinterpretation of results may occur if the information provided is inaccurate or incomplete. If results obtained do not match the clinical findings, additional testing should be considered.

Specific genetic events like copy number variants, translocations and repeat expansions may not be reliably detected with Exome Sequencing. In addition, due to limitations in technology, certain regions may either not be covered or may be poorly covered, where variants cannot be confidently detected.

Additional information

This test was developed and its performance validated by CENTOGENE AG. The US Food and Drug Administration (FDA) has determined that clearance or approval of this method is not necessary and thus neither have been obtained. This test has been developed for clinical purposes. All test results are reviewed, interpreted and reported by our scientific and medical experts.

In line with ACMG recommendations for reporting of incidental findings in clinical exome and genome sequencing (Genetics in Medicine, 2016), we report incidental findings, i.e. pathogenic variants (class 1) and likely pathogenic variants (class 2) only in the recommended genes for the recommended phenotypes.

To also exclude mistaken identity in your clinic, several guidelines recommend testing a second sample that is independently obtained from the proband. Please note that any further analysis will result in additional costs.

The classification of variants can change over the time. Please feel free to contact CENTOGENE (testing@centogene.com) in the future to determine if there have been any changes in classification of any reported variants.

Disclaimer

Any preparation and processing of a sample from patient material provided to CENTOGENE by a physician, clinical institute or a laboratory (by a "Partner") and the requested genetic and/or biochemical testing itself is based on the highest and most current scientific and analytical standards. However, in very few cases genetic or biochemical tests may not show the correct result, e.g. because of the quality of the material provided by a Partner to CENTOGENE or in cases where any test provided by CENTOGENE fails for unforeseeable or unknown reasons that cannot be influenced by CENTOGENE in advance. In such cases, CENTOGENE shall not be responsible and/or liable for the incomplete, potentially misleading or even wrong result of any testing if such issue could not be recognized by CENTOGENE in advance.

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